

Tugas Chapter 4

Analisis dan Pertanyaan Reflektif dan Praktik pengerjaan teknis

Analisis dan Pertanyaan Reflektif

Nama : Labiba Adinda Zahwana

Kelas : C

1. Apa perbedaan recovery dan remediation?

Recovery adalah proses mengembalikan sistem atau layanan agar kembali berjalan normal setelah terjadi insiden keamanan atau gangguan. Fokus utamanya adalah memulihkan operasional dan availability layanan secepat mungkin. Contohnya seperti restore backup, menjalankan ulang service, atau mengembalikan konfigurasi sistem yang rusak.

Sedangkan remediation adalah proses memperbaiki akar penyebab insiden agar masalah yang sama tidak terjadi kembali. Fokusnya berada pada peningkatan keamanan dan penghilangan celah yang menyebabkan insiden. Contohnya seperti patch vulnerability, memperbaiki konfigurasi keamanan, menghapus malware secara menyeluruh, menerapkan monitoring, atau memperketat access control.

Dalam praktik recovery lab yang dilakukan:

- a. Recovery : restore file aplikasi dari backup dan menjalankan kembali service hingga status menjadi RUNNING.
- b. Remediation : menghapus file mencurigakan `malware.sh`, melakukan validasi integritas backup, serta menerapkan kontrol keamanan tambahan agar insiden serupa tidak terulang.

2. Apa risiko jika recovery dilakukan tanpa validasi backup?

- Backup nya bisa korup atau rusak
- Backup ada kemungkinan sudah ada malware
- Recovery dapat gagal
- Sistem bisa mengalami kerusakan lebih besar
- Downtime layanan menjadi lama

Karena itu validasi checksum yang dilakukan pada praktik itu penting dilakukan sebelum restore.

3. Bagian mana dari proses ini yang termasuk DRP?

- Restore file dari backup

- Validasi backup
- Menjalankan ulang service
- Verifikasi layanan berjalan normal
- Dokumentasi recovery

Disaster recovery plan fokus pada pemulihan layanan dan infrastruktur setelah gangguan.

4. Apa dampak bisnis jika proses recovery gagal?

Dampak terhadap bisnis jika recovery process failed :

- Downtime layanan
- Kerugian finansial
- Kehilangan data
- Penurunan kepercayaan pelanggan
- Gangguan operasional perusahaan
- Risiko pelanggaran SLA
- Potensi reputasi buruk perusahaan

5. Kontrol keamanan apa yang seharusnya diterapkan sebelum insiden terjadi?

Kontrol keamanan yang sebaiknya diterapkan:

- Backup otomatis dan berkala
- Monitoring integritas file (AIDE/Wazuh)
- Least privilege access control
- Log monitoring dan SIEM
- Endpoint protection
- Immutable backup
- Patch management
- Incident response playbook
- Audit dan recovery drill berkala

Tugas Chapter 4

Praktik pengerjaan teknis

1. Laporan Teknis proses recovery

a. Identifikasi dampak insiden

Tahap awal dilakukan dengan pemeriksaan status service aplikasi menggunakan perintah:

```
labibaadinda@labibaadinda:~/recovery-lab-simple$ docker exec -it recovery-lab bash
root@4ff78d1f8fd1:/lab# service-control status
Service app status: FAILED / STOPPED
```

Hasil pemeriksaan menunjukkan bahwa service aplikasi berada dalam kondisi gagal berjalan.

Nah lanjut analisis log aplikasi menggunakan :

```
root@4ff78d1f8fd1:/lab# service-control logs
[2026-01-14 09:10:01] app: healthy heartbeat
[2026-01-14 09:10:06] app: healthy heartbeat
[2026-01-14 09:12:44] security: unauthorized file drop detected in /lab/app
[2026-01-14 09:13:02] app: startup failure - invalid bootstrap detected
[2026-01-14 09:13:03] app: process exited unexpectedly
```

Dari output diatas ditemukan insiden keamanan berupa :

- Unauthorized file drop
- Startup failure
- Process exited unexpectedly

Log menunjukkan adanya file mencurigakan pada direktori aplikasi yang menyebabkan aplikasi gagal melakukan bootstrap dan berhenti berjalan.

b. Containment atau isolasi sistem

Dilakukan investigasi terhadap direktori aplikasi:

```
root@4ff78d1f8fd1:/lab# cd /lab/app
root@4ff78d1f8fd1:/lab/app# ls -lah
total 28K
drwxrwxr-x 1 participant participant 4.0K May 10 02:01 .
drwxr-xr-x 1 participant participant 4.0K May 10 02:01 ..
-rwxrwxr-x 1 participant participant 258 May 10 01:59 app.sh
-rwxr-xr-x 1 root          root          58 May 10 02:01 malware.sh
-rwxrwxr-x 1 participant participant 142 May 10 02:01 start.sh
```

Terdapat file yang suspicious : malware.sh

Cek isi file malware.sh

```
root@4ff78d1f8fd1:/lab/app# cat malware.sh
#!/usr/bin/env bash
echo "simulated persistence artifact"
root@4ff78d1f8fd1:/lab/app# rm malware.sh
```

File tersebut diidentifikasi sebagai artefak persistence yang tidak legitimate

Tindakan containment dilakukan dengan menghapus file tsb:

```
root@4ff78d1f8fd1:/lab/app# rm malware.sh
```

c. Validasi backup

Dilakukan pemeriksaan direktori backup dan dilakukan verifikasi integritas backup menggunakan checksum

```
root@4ff78d1f8fd1:/lab/app# ls -lh /lab/backup
total 8.0K
-rw-r--r-- 1 root root 96 May 10 02:01 app_backup.sha256
-rw-r--r-- 1 root root 353 May 10 02:01 app_backup.tar.gz
root@4ff78d1f8fd1:/lab/app# sha256sum -c /lab/backup/app_backup.sha256
/lab/backup/app_backup.tar.gz: OK
root@4ff78d1f8fd1:/lab/app# cd /lab
root@4ff78d1f8fd1:/lab# tar -xvzf backup/app_backup.tar.gz
app/
app/app.sh
app/start.sh
```

Hal ini menunjukkan backup valid dan aman digunakan untuk proses recovery.

d. Recovery sistem

```
root@4ff78d1f8fd1:/lab/app# cd /lab
root@4ff78d1f8fd1:/lab# tar -xvzf backup/app_backup.tar.gz
app/
app/app.sh
app/start.sh
```

File app berhasil di pulihkan

Setelah restore selesai, service dijalankan kembali

```
root@4ff78d1f8fd1:/lab# ls -lah /lab/app
total 24K
drwxr-xr-x 1 root      root      4.0K May 10 02:01 .
drwxr-xr-x 1 participant participant 4.0K May 10 02:01 ..
-rwxr-xr-x 1 root      root      258 May 10 02:01 app.sh
-rwxr-xr-x 1 root      root      59 May 10 02:01 start.sh
root@4ff78d1f8fd1:/lab# service-control start
Service app started successfully with PID 51
root@4ff78d1f8fd1:/lab# service-control status
Service app status: RUNNING (PID 51)
```

Hal ini menunjukkan proses recovery berhasil mengembalikan aplikasi ke kondisi operasional normal.

e. Post-incident validation

Pemeriksaan log pasca recover dilakukan menggunakan :

```
root@4ff78d1f8fd1:/lab# service-control logs
[2026-01-14 09:10:01] app: healthy heartbeat
[2026-01-14 09:10:06] app: healthy heartbeat
[2026-01-14 09:12:44] security: unauthorized file drop detected in /lab/app
[2026-01-14 09:13:02] app: startup failure - invalid bootstrap detected
[2026-01-14 09:13:03] app: process exited unexpectedly
[2026-05-10 02:09:25] app: starting web payroll helper
[2026-05-10 02:09:25] app: healthy heartbeat
[2026-05-10 02:09:31] app: healthy heartbeat
[2026-05-10 02:09:36] app: healthy heartbeat
[2026-05-10 02:09:41] app: healthy heartbeat
[2026-05-10 02:09:46] app: healthy heartbeat
[2026-05-10 02:09:51] app: healthy heartbeat
[2026-05-10 02:09:56] app: healthy heartbeat
```

dan cat app.log nya

```
root@4ff78d1f8fd1:/lab# cat /lab/logs/app.log
[2026-01-14 09:10:01] app: healthy heartbeat
[2026-01-14 09:10:06] app: healthy heartbeat
[2026-01-14 09:12:44] security: unauthorized file drop detected in /lab/app
[2026-01-14 09:13:02] app: startup failure - invalid bootstrap detected
[2026-01-14 09:13:03] app: process exited unexpectedly
[2026-05-10 02:09:25] app: starting web payroll helper
[2026-05-10 02:09:25] app: healthy heartbeat
[2026-05-10 02:09:31] app: healthy heartbeat
[2026-05-10 02:09:36] app: healthy heartbeat
[2026-05-10 02:09:41] app: healthy heartbeat
[2026-05-10 02:09:46] app: healthy heartbeat
[2026-05-10 02:09:51] app: healthy heartbeat
[2026-05-10 02:09:56] app: healthy heartbeat
[2026-05-10 02:10:01] app: healthy heartbeat
[2026-05-10 02:10:06] app: healthy heartbeat
[2026-05-10 02:10:11] app: healthy heartbeat
[2026-05-10 02:10:16] app: healthy heartbeat
```

Tidak ditemukan lagi error ataupun aktivitas yang mencurigakan.

```
root@4ff78d1f8fd1:/lab# ls -lah /lab/app
total 24K
drwxr-xr-x 1 root      root      4.0K May 10 02:01 .
drwxr-xr-x 1 participant participant 4.0K May 10 02:01 ..
-rwxr-xr-x 1 root      root      258 May 10 02:01 app.sh
-rwxr-xr-x 1 root      root      59 May 10 02:01 start.sh
```

File mencurigakan malware.sh sudah tidak ditemukan.

Validasi otomatis dilakukan menggunakan checklab

```
root@4ff78d1f8fd1:/lab# check-lab
[PASS] Backup checksum valid
[PASS] Suspicious file removed
[PASS] start.sh restored correctly
[PASS] Service is running
[PASS] Application produced healthy log entries

RESULT: LAB RECOVERY SUCCESS (5/5)
```

2. Timeline insiden dan pemulihan

- Timeline insiden

```
labibaadinda@labibaadinda:~/recovery-lab-simple$ docker exec -it recovery-lab bash
root@4fff78d1f8fd1:/lab# service-control status
Service app status: FAILED / STOPPED
root@4fff78d1f8fd1:/lab# service-control logs
[2026-01-14 09:10:01] app: healthy heartbeat
[2026-01-14 09:10:06] app: healthy heartbeat
[2026-01-14 09:12:44] security: unauthorized file drop detected in /lab/app
[2026-01-14 09:13:02] app: startup failure - invalid bootstrap detected
[2026-01-14 09:13:03] app: process exited unexpectedly
```

Tahap	Aktivitas	Bukti
09:10:01	Aplikasi berjalan normal	healthy heartbeat
09:12:44	Terdeteksi unauthorized file drop	security log
09:13:02	Startup failure terjadi	invalid bootstrap detected
09:13:03	Service berhenti	process exited unexpectedly
Recovery-01	Pemeriksaan status service	service-control status
Recovery-02	Analisis log aplikasi	service-control logs
Recovery-03	Investigasi direktori aplikasi	ls -lah /lab/app
Recovery-04	Identifikasi file malware.sh	cat malware.sh
Recovery-05	Penghapusan file mencurigakan	rm malware.sh
Recovery-06	Pemeriksaan backup	ls -lh /lab/backup
Recovery-07	Verifikasi checksum backup	sha256sum -c
Recovery-08	Restore aplikasi dari backup	tar -xvzf
Recovery-09	Menjalankan ulang service	service-control start
Recovery-10	Validasi service running	service-control status
Recovery-11	Analisis log pasca recovery	cat /lab/logs/app.log
Recovery-12	Validasi otomatis recovery	check-lab
Final	Recovery berhasil	LAB RECOVERY SUCCESS (5/5)

3. Bukti eksekusi perintah (screenshot)

```
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; p
   Active: active (running) since Sun 2026-05-10 08:41:18 WIB; 18min >
   Invocation: 6ce7902927fc4bbda187e03b81545688
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 1799 (dockerd)
      Tasks: 17
     Memory: 134.5M (peak: 137.2M)
        CPU: 4.221s
    CGroup: /system.slice/docker.service
            └─1799 /usr/bin/dockerd -H fd:// --containerd=/run/contain>

May 10 08:41:17 labibaadinda dockerd[1799]: time="2026-05-10T08:41:17.8>
May 10 08:41:17 labibaadinda dockerd[1799]: time="2026-05-10T08:41:17.8>
May 10 08:41:17 labibaadinda dockerd[1799]: time="2026-05-10T08:41:17.8>
May 10 08:41:17 labibaadinda dockerd[1799]: time="2026-05-10T08:41:17.8>
May 10 08:41:18 labibaadinda dockerd[1799]: time="2026-05-10T08:41:18.1>
May 10 08:41:18 labibaadinda dockerd[1799]: time="2026-05-10T08:41:18.1>
May 10 08:41:18 labibaadinda dockerd[1799]: time="2026-05-10T08:41:18.5>
May 10 08:41:18 labibaadinda dockerd[1799]: time="2026-05-10T08:41:18.5>
May 10 08:41:18 labibaadinda dockerd[1799]: time="2026-05-10T08:41:18.5>
May 10 08:41:18 labibaadinda systemd[1]: Started docker.service - Docke>
Docker version 29.3.1, build c2be9cc
Unable to find image 'hello-world:latest' locally
labibaadinda@labibaadinda:~$ git clone https://github.com/isanimal/recovery-lab-simple
Cloning into 'recovery-lab-simple'...
remote: Enumerating objects: 26, done.
remote: Counting objects: 100% (26/26), done.
remote: Compressing objects: 100% (25/25), done.
remote: Total 26 (delta 6), reused 15 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (26/26), 6.89 KiB | 1.38 MiB/s, done.
Resolving deltas: 100% (6/6), done.
labibaadinda@labibaadinda:~$ cd recovery-lab-simple
labibaadinda@labibaadinda:~/recovery-lab-simple$ ls
Dockerfile  docker-compose.yml  scripts
README.md   lab
labibaadinda@labibaadinda:~/recovery-lab-simple$ docker compose up -d --build
[+] Building 65.0s (14/14) FINISHED
=> [internal] load local bake definitions          0.0s
=> reading from stdin 5768                        0.0s

labibaadinda@labibaadinda:~/recovery-lab-simple$ docker compose up -d --build
[+] Building 65.0s (14/14) FINISHED
=> [internal] load local bake definitions          0.0s
=> reading from stdin 5768                        0.0s
=> [internal] load build definition from Dockerfile 0.3s
=> transferring dockerfile: 612B                  0.0s
=> [internal] load metadata for docker.io/library/ubuntu:22.04 3.5s
[auth] library/ubuntu:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore                   0.1s
=> transferring context: 2B                        0.0s
=> [1/6] FROM docker.io/library/ubuntu:22.04@sha256:962f6cadeae 15.7s
=> resolve docker.io/library/ubuntu:22.04@sha256:962f6cadeae0 0.1s
=> sha256:f63eb04151bcac21ad049f8d781b97b2 29.74MB / 29.74MB 12.9s
=> extracting sha256:f63eb04151bcac21ad049f8d781b97b219aba392 2.6s
=> [internal] load build context                   0.1s
=> transferring context: 6.69kB                   0.0s
=> [2/6] RUN apt-get update && apt-get install -y  bash 37.8s
=> [3/6] WORKDIR /lab                             0.1s
=> [4/6] COPY lab/ /lab/                          0.1s
=> [5/6] COPY scripts/ /usr/local/bin/            0.1s
=> [6/6] RUN chmod +x /usr/local/bin/* && useradd -m -s /bin 0.6s
=> exporting to image                             5.5s
=> exporting layers                               3.8s
=> exporting manifest sha256:10070ae4e04c4a5ade63e4acfc842f2a 0.0s
=> exporting config sha256:285f24d4ef696e93498048ae9a09e5bf38 0.0s
=> exporting attestation manifest sha256:b62ded28ed5bdb099199 0.1s
=> exporting manifest list sha256:d46b9a1369685b8e2ac6e282409 0.0s
=> naming to docker.io/library/recovery-lab-simple-recovery-l 0.0s
=> unpacking to docker.io/library/recovery-lab-simple-recover 1.4s
=> resolving provenance for metadata file          0.0s
[+] up 3/3
✔Image recovery-lab-simple-recovery-lab Built 65.1s
✔Network recovery-lab-simple-default Created 0.2s
✔Container recovery-lab Started 1.1s
labibaadinda@labibaadinda:~/recovery-lab-simple$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
4ff78d1f8fd1  recovery-lab-simple-recovery-lab  "/usr/local/bin/lab-..." 9 seconds ago  Up 9 seconds  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tc
p_recovery-lab
labibaadinda@labibaadinda:~/recovery-lab-simple$ docker exec -it recovery-lab bash
root@4ff78d1f8fd1:/lab# service-control status
Service app status: FAILED / STOPPED
root@4ff78d1f8fd1:/lab# service-control logs
```

```

labibaadinda@labibaadinda:~/recovery-lab-simple$ docker exec -it recovery-lab bash
root@4fff78d1f8fd1:/lab# service-control status
Service app status: FAILED / STOPPED
root@4fff78d1f8fd1:/lab# service-control logs
[2026-01-14 09:10:01] app: healthy heartbeat
[2026-01-14 09:10:06] app: healthy heartbeat
[2026-01-14 09:12:44] security: unauthorized file drop detected in /lab/app
[2026-01-14 09:13:02] app: startup failure - invalid bootstrap detected
[2026-01-14 09:13:03] app: process exited unexpectedly
root@4fff78d1f8fd1:/lab# cd /lab/app
root@4fff78d1f8fd1:/lab/app# ls -lah
total 28K
drwxrwxr-x 1 participant participant 4.0K May 10 02:01 .
drwxr-xr-x 1 participant participant 4.0K May 10 02:01 ..
-rwxrwxr-x 1 participant participant 258 May 10 01:59 app.sh
-rwxr-xr-x 1 root root 58 May 10 02:01 malware.sh
-rwxrwxr-x 1 participant participant 142 May 10 02:01 start.sh
root@4fff78d1f8fd1:/lab/app# cat malware.sh
#!/usr/bin/env bash
echo "simulated persistence artifact"
root@4fff78d1f8fd1:/lab/app# rm malware.sh
root@4fff78d1f8fd1:/lab/app# ls -lh /lab/backup
total 8.0K
-rw-r--r-- 1 root root 96 May 10 02:01 app_backup.sha256
-rw-r--r-- 1 root root 353 May 10 02:01 app_backup.tar.gz
root@4fff78d1f8fd1:/lab/app# sha256sum -c /lab/backup/app_backup.sha256
/lab/backup/app_backup.tar.gz: OK
root@4fff78d1f8fd1:/lab/app# cd /lab
root@4fff78d1f8fd1:/lab# tar -xvzf backup/app_backup.tar.gz
app/
app/app.sh
app/start.sh
root@4fff78d1f8fd1:/lab# ls -lah /lab/app
total 24K
drwxr-xr-x 1 root root 4.0K May 10 02:01 .
drwxr-xr-x 1 participant participant 4.0K May 10 02:01 ..
-rwxr-xr-x 1 root root 258 May 10 02:01 app.sh
-rwxr-xr-x 1 root root 59 May 10 02:01 start.sh
root@4fff78d1f8fd1:/lab# service-control start
Service app started successfully with PID 51
root@4fff78d1f8fd1:/lab# service-control status
Service app status: RUNNING (PID 51)
root@4fff78d1f8fd1:/lab# service-control logs

```



```
root@4ff78d1f8fd1:/lab# service-control logs
[2026-01-14 09:10:01] app: healthy heartbeat
[2026-01-14 09:10:06] app: healthy heartbeat
[2026-01-14 09:12:44] security: unauthorized file drop detected in /lab/app
[2026-01-14 09:13:02] app: startup failure - invalid bootstrap detected
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[2026-05-10 02:09:25] app: starting web payroll helper
[2026-05-10 02:09:25] app: healthy heartbeat
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[2026-05-10 02:09:36] app: healthy heartbeat
[2026-05-10 02:09:41] app: healthy heartbeat
[2026-05-10 02:09:46] app: healthy heartbeat
[2026-05-10 02:09:51] app: healthy heartbeat
[2026-05-10 02:09:56] app: healthy heartbeat
root@4ff78d1f8fd1:/lab# cat /lab/logs/app.log
[2026-01-14 09:10:01] app: healthy heartbeat
[2026-01-14 09:10:06] app: healthy heartbeat
[2026-01-14 09:12:44] security: unauthorized file drop detected in /lab/app
[2026-01-14 09:13:02] app: startup failure - invalid bootstrap detected
[2026-01-14 09:13:03] app: process exited unexpectedly
[2026-05-10 02:09:25] app: starting web payroll helper
[2026-05-10 02:09:25] app: healthy heartbeat
[2026-05-10 02:09:31] app: healthy heartbeat
[2026-05-10 02:09:36] app: healthy heartbeat
[2026-05-10 02:09:41] app: healthy heartbeat
[2026-05-10 02:09:46] app: healthy heartbeat
[2026-05-10 02:09:51] app: healthy heartbeat
[2026-05-10 02:09:56] app: healthy heartbeat
[2026-05-10 02:10:01] app: healthy heartbeat
[2026-05-10 02:10:06] app: healthy heartbeat
[2026-05-10 02:10:11] app: healthy heartbeat
[2026-05-10 02:10:16] app: healthy heartbeat
root@4ff78d1f8fd1:/lab# ls -lah /lab/app
total 24K
drwxr-xr-x 1 root      root      4.0K May 10 02:01 .
drwxr-xr-x 1 participant participant 4.0K May 10 02:01 ..
-rwxr-xr-x 1 root      root       258 May 10 02:01 app.sh
-rwxr-xr-x 1 root      root        59 May 10 02:01 start.sh
```

```
root@4ff78d1f8fd1:/lab# ls -lah /lab/app
total 24K
drwxr-xr-x 1 root      root      4.0K May 10 02:01 .
drwxr-xr-x 1 participant participant 4.0K May 10 02:01 ..
-rwxr-xr-x 1 root      root       258 May 10 02:01 app.sh
-rwxr-xr-x 1 root      root        59 May 10 02:01 start.sh
root@4ff78d1f8fd1:/lab# check-lab
[PASS] Backup checksum valid
[PASS] Suspicious file removed
[PASS] start.sh restored correctly
[PASS] Service is running
[PASS] Application produced healthy log entries

RESULT: LAB RECOVERY SUCCESS (5/5)
```